

willing to pay for the privilege to trade as soon as they desire. As investors compete within an asset class, they covet and pay up for liquidity.

4.3. MARKET MAKING

A market maker supplies liquidity by acting as an intermediary between buyers and sellers.³² Liquidity provision is costly. Market makers need capital to withstand a potential onslaught of buy or sell orders, and at any time they can be transacting with investors who have superior information. In compensation for these costs, market makers buy at low prices and sell at prices around “fair value.” Investors transacting with the market maker pay the *bid-ask spread*.

In liquid stock and bond markets, market making is now synonymous with high frequency trading by investors who build massive computer infrastructure to submit buy and sell orders within fractions of a second. More than 70% of dollar trading volume on U.S. equity exchanges is believed due to high frequency traders.³³ Many successful hedge funds specialize in high frequency trading (see chapter 17).

Many asset owners cannot collect illiquidity risk premiums by building high-frequency trading systems, nor would they wish to enter this business (directly or indirectly). But there is a way large asset owners can do a low-frequency version of market making.

Dimensional Funds Advisors (DFA) is a funds management company that started in 1981 by specializing in small-cap equities. DFA’s strategies have deep roots in academic factor models, and its founders, David Booth and Rex Sinquefeld, roped in the big guns of the finance literature, Fama, French, and others, in building the company. From the start, DFA positioned itself as a liquidity provider of small stocks, and market making was an integral part of its investment strategy.³⁴ When other investors seek to urgently offload large amounts of small stocks, DFA takes the other side and buys at a discount. Similarly, DFA offers small-cap equities at a premium to investors who demand immediate liquidity.

Large asset owners, like sovereign wealth funds and large pension funds, are in a position to act as liquidity providers, especially in more illiquid markets. They can accept large blocks of bonds, shares, or even portfolios of property at discount and sell these large blocks at premiums. They can do this by calculating limits within their (benchmark tracking error) constraints on how much they are willing to transact. That is, they can provide liquidity in different securities up to a certain amount so that they do not stray too far from their benchmarks. Buyers and sellers will come to them as they develop reputations for providing liquidity.

³² O’Hara (1995) provides a summary of theoretical models of market making.

³³ See Zhang (2010).

³⁴ See Keim (1999) and the Harvard Business School case study, Dimensional Fund Advisors, 2002, written by Randolph Cohen.